

# CUTE-YOLO Tester Guide

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# What is CUTE YOLO

CUTE-YOLO is a compact, single-class object detector for microcontrollers.

The micro:bit V2 runs the INT8 model using Noodle and  $\mu$ T-Kernel 3.0. The computer sends a  $128 \times 128$  grayscale image; the device returns detection boxes.

A .cute package contains the model and its configuration. H is the number of hybrid blocks; M is the number of hidden head layers.

# Step 1

Connect the supplied micro:bit V2 to the computer with a USB data cable.

Its  $\mu$ T-Kernel detector firmware is already installed. Select the micro:bit serial port in the tester; use Refresh ports if it is not listed.

The shipped board already have a face model. Click Read active model to check its label, H/M configuration, and flash slot.

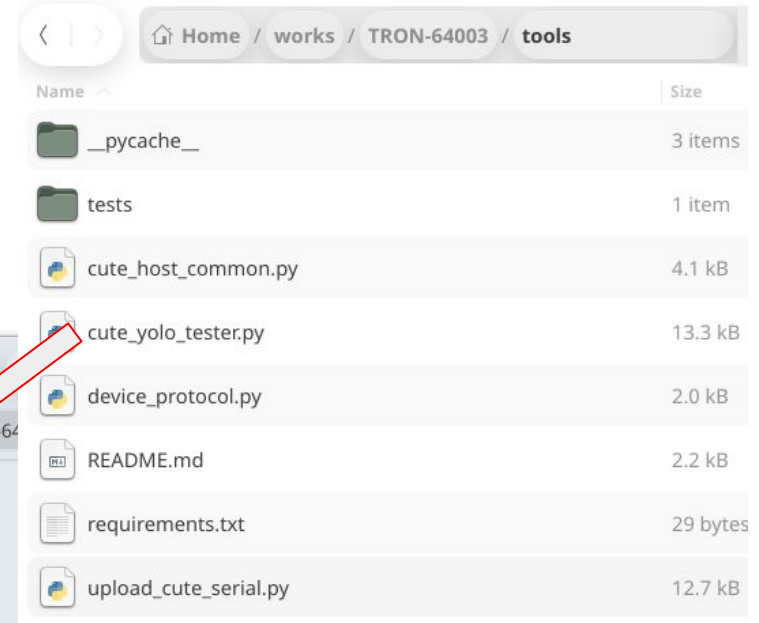
# Step 1

Run from the tools folder:

```
python cute_yolo_tester.py
```



A terminal window titled "auralius@auralius-x270: ~/works/TRON-64003/tools" showing the command `python ./cute_yolo_tester.py` being executed. The prompt is `(py.3.12.4) auralius@auralius-x270:~/works/TRON-64003/tools$`. A red arrow points from the file `cute_yolo_tester.py` in the file explorer to the command in the terminal.



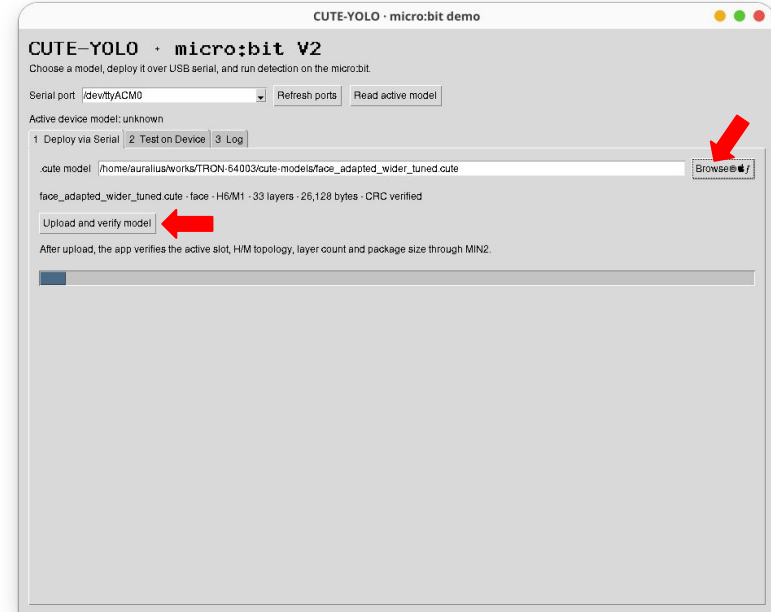
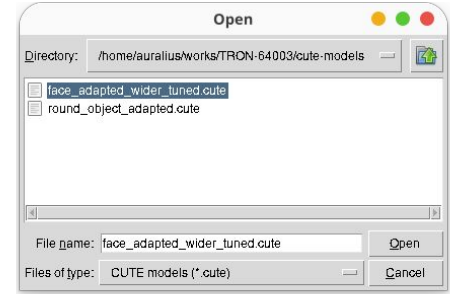
## Step 2

Flash the CUTE model to the microbit.

We provided to CUTE models: round object detection and face detection.

The shipped Microbit is already flashed with CUTE model for face detection.

We can skip this step if we just want to test the detection.

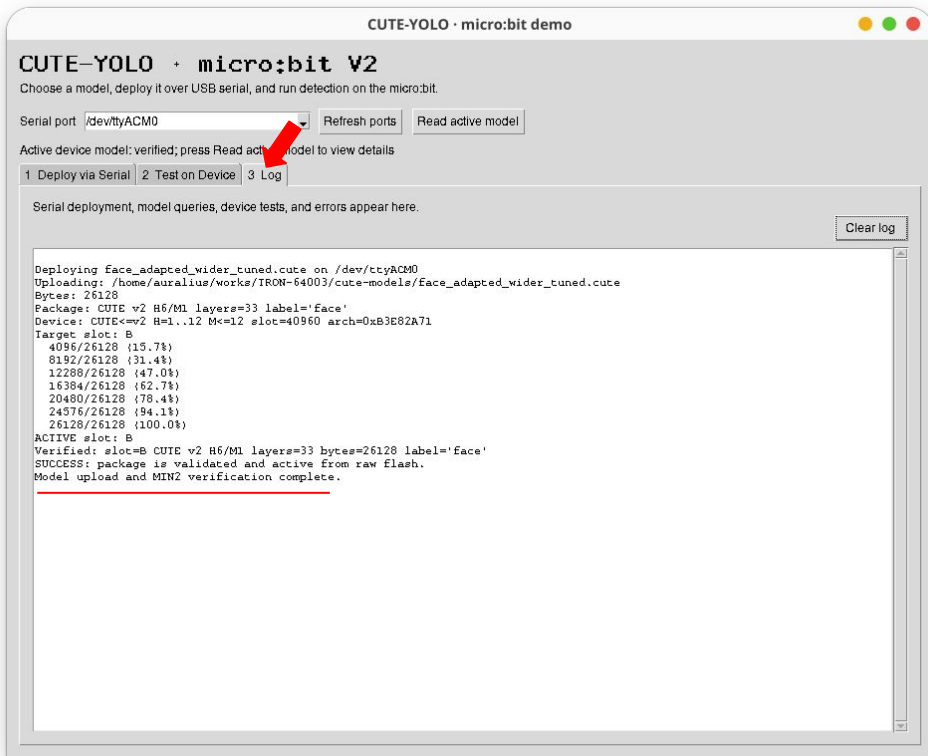


## Step 2

Open the Log tab after deployment.

Look for ACTIVE slot and the final MIN2 verification. The reported label and H/M values should match the selected .cute file.

If an error appears, check the USB serial port and close any other application using it.



# Step 3

Open Test on Device. Select an image or start the laptop webcam.

Click Capture and detect on device. One  $128 \times 128$  grayscale frame is sent over USB serial.

The micro:bit returns boxes and timing for display in the tester.

CUTE-YOLO · micro:bit demo

### CUTE-YOLO + micro:bit V2

Choose a model, deploy it over USB serial, and run detection on the micro:bit.

Serial port:

Active model: face - H6/M1 - slot B

1 Deploy via Serial 2 Test on Device 3 Log

Source (red square = detector crop)



128x128 device input and detections



4 detection(s) - inference 15310 ms - total 16790 ms

| confidence | x1 | y1 | x2 | y2 |
|------------|----|----|----|----|
| 0.937      | 45 | 52 | 54 | 62 |
| 0.839      | 71 | 59 | 80 | 70 |
| 0.812      | 84 | 35 | 93 | 46 |
| 0.702      | 55 | 54 | 63 | 64 |

## Step 4

Check the resulting image on the display.



## If the LCD stays white

The attached LCD may stay white after power-up despite our startup mitigations. This is a known limitation of the current demo hardware.

Briefly press and release Reset, then wait a few seconds. If it stays white, try one or two more brief presses.

Do not hold Reset: a long press puts micro:bit V2 to sleep.

The test can still run with a blank LCD. If the tester receives boxes and timing, continue using the computer display.

## If the serial test does not start

No serial port: reconnect the USB data cable and click Refresh ports.

Port busy or timeout: close another serial monitor or tester, then retry.

No active model: deploy a compatible .cute file and check MIN2 in the Log tab.

Unexpected boxes: check the active model label. A face model should be tested with faces; a round-object model with round objects.

